

Sri Lanka Institute of Information Technology



SE3100 : Architecture based Development

Lab Sheet No: 05

Student ID	IT24102503
Student Name	Herath H.M.P.S.R.K
Campus/ Center name	Kandy Uni
Specialization	Software Engineering

```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

PS C:\Users\User> cd Downloads\dashmarket-lab
PS C:\Users\User\Downloads\dashmarket-lab> docker swarm init
Swarm initialized: current node (iafeekqtl3hxzoolw84t0siuq) is now a manager
.

To add a worker to this swarm, run the following command:

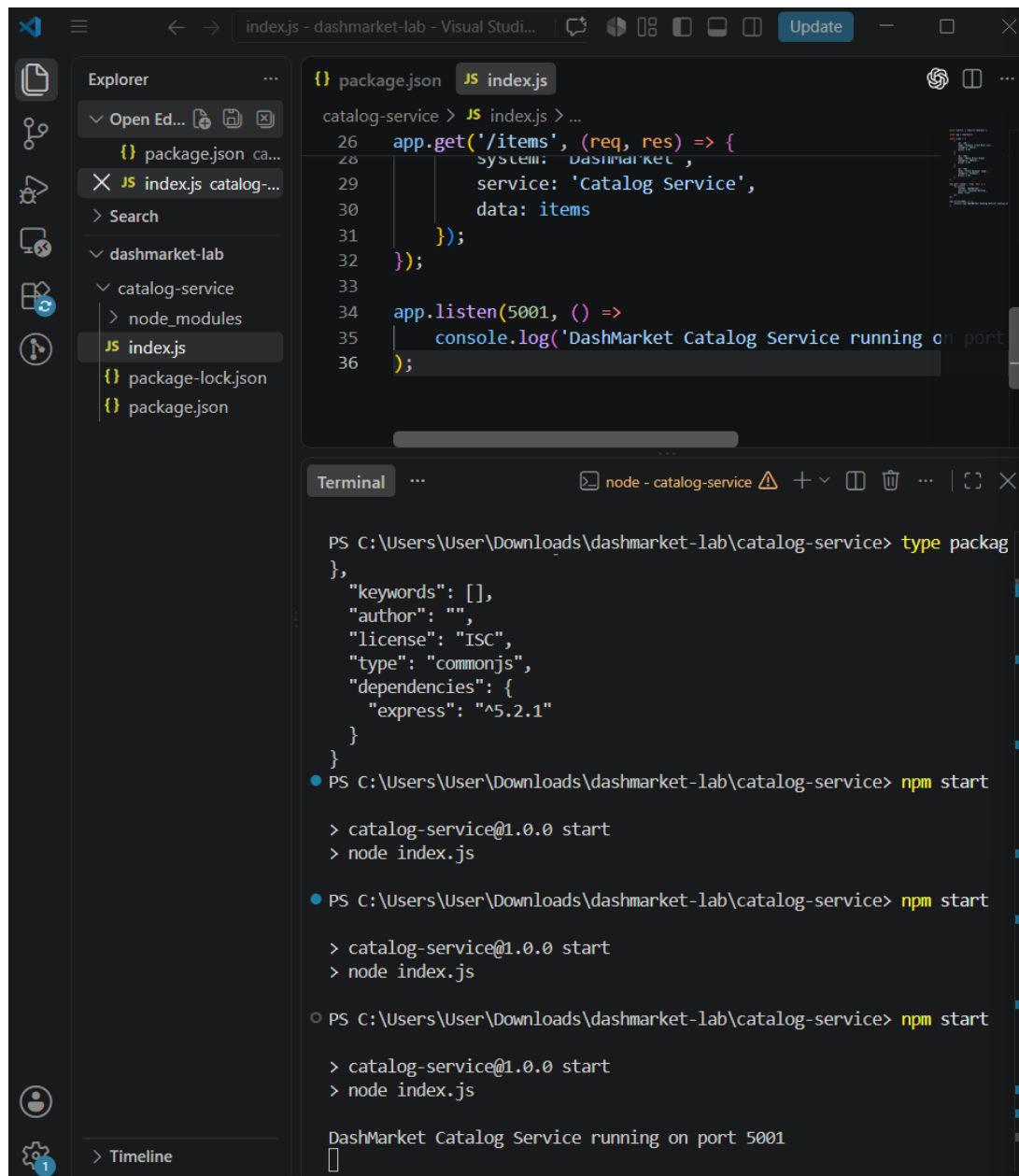
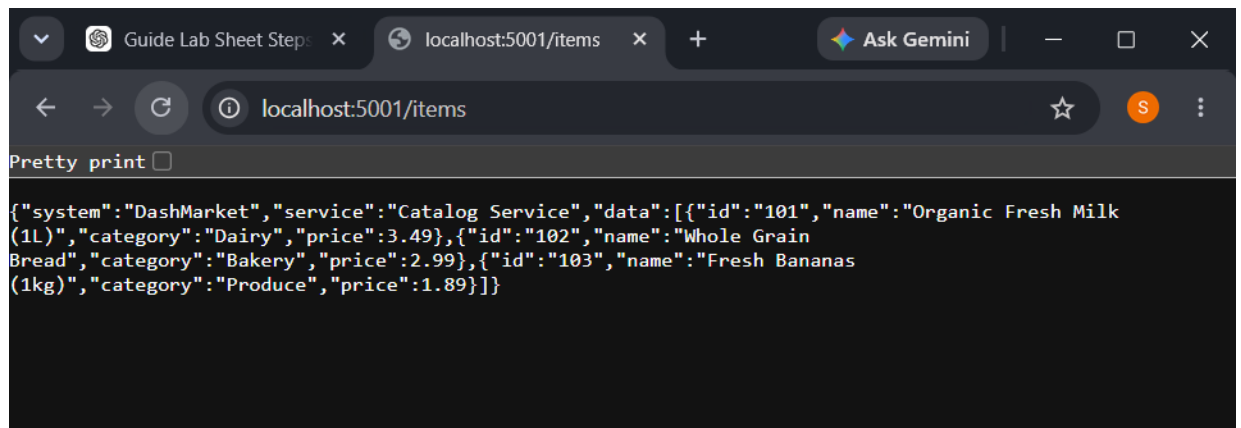
    docker swarm join --token SWMTKN-1-51pkv0hv4vy35ibuotp339961gs2gr2ac8jgj
9mgctx4c3n7ti-2yzbvgwj3fflvje3xft0d6j57 192.168.65.3:2377

To add a manager to this swarm, run 'docker swarm join-token manager' and fo
llow the instructions.
PS C:\Users\User\Downloads\dashmarket-lab> |
```

```
Restricted Mode is intended for safe code browsing. Trust this folder to enable all features. Manage Learn More

Explorer
  Open Edit... 1 unsaved
  {} package.json... 1
  Search
  dashmarket-lab
    catalog-service
      node_modules
        package-lock.json
        package.json 1

{} package.json 1
catalog-service > {} package.json > ...
1  {
2    "name": "catalog-service",
3    "version": "1.0.0",
4    "description": "",
5    "main": "index.js",
6    "scripts": {
7      "start": "node index.js"
8    },
9    "keywords": [],
10   "author": "",
11   "license": "ISC",
12   "type": "commonjs",
13   "dependencies": {
14     "express": "^5.2.1"
15   }
16 }
17
```



```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

PS C:\Users\User> cd Downloads\dashmarket-lab
PS C:\Users\User\Downloads\dashmarket-lab> mkdir notification-service

Directory: C:\Users\User\Downloads\dashmarket-lab

Mode                LastWriteTime         Length Name
----                -
d-----          8/23/2026  10:30 PM              notification-service

PS C:\Users\User\Downloads\dashmarket-lab> cd notification-service
PS C:\Users\User\Downloads\dashmarket-lab\notification-service> npm init -y
Wrote to C:\Users\User\Downloads\dashmarket-lab\notification-service\package
.json:

{
  "name": "notification-service",
  "version": "1.0.0",
  "description": "",
  "main": "index.js",
  "scripts": {
    "test": "echo \"Error: no test specified\" && exit 1"
  },
  "keywords": [],
  "author": "",
  "license": "ISC",
  "type": "commonjs"
}

PS C:\Users\User\Downloads\dashmarket-lab\notification-service> npm install
express

added 68 packages, and audited 69 packages in 2s

27 packages are looking for funding
  run `npm fund` for details

found 0 vulnerabilities
PS C:\Users\User\Downloads\dashmarket-lab\notification-service> |
```

```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

PS C:\Users\User> cd Downloads\dashmarket-lab
PS C:\Users\User\Downloads\dashmarket-lab> mkdir delivery-service

Directory: C:\Users\User\Downloads\dashmarket-lab

Mode                LastWriteTime         Length Name
----                -
d-----            8/23/2026 10:29 PM                delivery-service

PS C:\Users\User\Downloads\dashmarket-lab> cd delivery-service
PS C:\Users\User\Downloads\dashmarket-lab\delivery-service> npm init -y
Wrote to C:\Users\User\Downloads\dashmarket-lab\delivery-service\package.js
on:
{
  "name": "delivery-service",
  "version": "1.0.0",
  "description": "",
  "main": "index.js",
  "scripts": {
    "test": "echo \"Error: no test specified\" && exit 1"
  },
  "keywords": [],
  "author": "",
  "license": "ISC",
  "type": "commonjs"
}

PS C:\Users\User\Downloads\dashmarket-lab\delivery-service> npm install exp
ress

added 68 packages, and audited 69 packages in 3s

27 packages are looking for funding
  run `npm fund` for details

found 0 vulnerabilities
PS C:\Users\User\Downloads\dashmarket-lab\delivery-service> |
```

```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

PS C:\Users\User> cd Downloads\dashmarket-lab
PS C:\Users\User\Downloads\dashmarket-lab> mkdir ^C
PS C:\Users\User\Downloads\dashmarket-lab> mkdir order-service

Directory: C:\Users\User\Downloads\dashmarket-lab

Mode                LastWriteTime         Length Name
----                -
d-----           8/23/2026 10:27 PM              order-service

PS C:\Users\User\Downloads\dashmarket-lab> cd order-service
PS C:\Users\User\Downloads\dashmarket-lab\order-service>
PS C:\Users\User\Downloads\dashmarket-lab\order-service> npm init -y
Wrote to C:\Users\User\Downloads\dashmarket-lab\order-service\package.json:

{
  "name": "order-service",
  "version": "1.0.0",
  "description": "",
  "main": "index.js",
  "scripts": {
    "test": "echo \"Error: no test specified\" && exit 1"
  },
  "keywords": [],
  "author": "",
  "license": "ISC",
  "type": "commonjs"
}

PS C:\Users\User\Downloads\dashmarket-lab\order-service> npm install express

added 68 packages, and audited 69 packages in 4s

27 packages are looking for funding
  run `npm fund` for details

found 0 vulnerabilities
PS C:\Users\User\Downloads\dashmarket-lab\order-service> |
```

index.js - dashmarket-lab - Visual Studi...

Update

Explorer

> Open Editors

> Search

das...

catalog-service

node_modules

Dockerfile

JS index.js

package-lock.json

package.json

delivery-service

node_modules

Dockerfile

JS index.js

package-lock.json

package.json

notification-service

node_modules

Dockerfile

JS index.js

package-lock.json

package.json

order-service

node_modules

Dockerfile

JS index.js

package-lock.json

package.json

docker-compose.yml

nginx.conf

Timeline

notification-service {} package.json delivery-service JS index.js delivery-service

delivery-service > JS index.js > ...

```
5  app.get('/:id', (req, res) => {
14  });
15  });
16
17  app.listen(5003, () =>
18  console.log('DashMarket Delivery Service running on port
19  );
```

Terminal

node - catalog-service

```
PS C:\Users\User\Downloads\dashmarket-lab\catalog-service> type packag
},
"keywords": [],
"author": "",
"license": "ISC",
"type": "commonjs",
"dependencies": {
  "express": "^5.2.1"
}
}
● PS C:\Users\User\Downloads\dashmarket-lab\catalog-service> npm start

> catalog-service@1.0.0 start
> node index.js

● PS C:\Users\User\Downloads\dashmarket-lab\catalog-service> npm start

> catalog-service@1.0.0 start
> node index.js

○ PS C:\Users\User\Downloads\dashmarket-lab\catalog-service> npm start

> catalog-service@1.0.0 start
> node index.js

DashMarket Catalog Service running on port 5001
█
```

Launchpad

0 0

Screen Reader Optimized

Ln 19, Col 3

Spaces: 4

UTF-8

CRLF

{ } JavaScript

```
Guide Lab Sheet Steps x localhost:5001/items x + Ask Gemini
localhost:5001/items
Pretty print
{"system":"DashMarket","service":"Catalog Service","data":[{"id":"101","name":"Organic Fresh Milk (1L)","category":"Dairy","price":3.49}, {"id":"102","name":"Whole Grain Bread","category":"Bakery","price":2.99}, {"id":"103","name":"Fresh Bananas (1kg)","category":"Produce","price":1.89}]}
```

```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

PS C:\Users\User> cd Downloads\dashmarket-lab
PS C:\Users\User\Downloads\dashmarket-lab> docker build -t dashmarket-catalog:1.0 ./catalog-service
[+] Building 75.6s (11/11) FINISHED                                docker:desktop-linux
=> [internal] load build definition from Dockerfile                0.1s
=> => transferring dockerfile: 170B                                0.0s
=> [internal] load metadata for docker.io/library/node:lts-alpine 4.6s
=> [auth] library/node:pull token for registry-1.docker.io        0.0s
=> [internal] load .dockerignore                                   0.0s
=> => transferring context: 2B                                       0.0s
=> [1/5] FROM docker.io/library/node:lts-alpine@sha256:d32cdf619f63fe0471182d08 66.7s
=> => resolve docker.io/library/node:lts-alpine@sha256:d32cdf619f63fe0471182d089 0.0s
=> => sha256:e27d3f3e3cb67c9f573672ff4d772cac332d144c55b1f4cf92de199 446B / 446B 0.5s
=> => sha256:4a7016f72779c9040cee0836ca5b79ae36c19d053dff298915d 1.26MB / 1.26MB 2.2s
=> => sha256:08d42cdba493bec7b21d91857a92221fcd480ebfe48f8da3 53.10MB / 53.10MB 65.3s
=> => sha256:55afa1ecc21d2bb5e5045f32dafee56272ffdd89860bac26f6c 3.85MB / 3.85MB 12.5s
=> => extracting sha256:55afa1ecc21d2bb5e5045f32dafee56272ffdd89860bac26f6c321234 0.2s
=> => extracting sha256:08d42cdba493bec7b21d91857a92221fcd480ebfe48f8da3bbd55721 1.1s
=> => extracting sha256:4a7016f72779c9040cee0836ca5b79ae36c19d053dff298915dbb192 0.0s
=> => extracting sha256:e27d3f3e3cb67c9f573672ff4d772cac332d144c55b1f4cf92de199f 0.0s
=> [internal] load build context                                    3.4s
=> => transferring context: 2.30MB                                    3.4s
=> [2/5] WORKDIR /app                                              0.1s
=> [3/5] COPY package*.json ./                                     0.0s
=> [4/5] RUN npm install                                           3.0s
=> [5/5] COPY . .                                                  0.1s
=> exporting to image                                              0.9s
=> => exporting layers                                              0.3s
=> => exporting manifest sha256:a25de5ad127f431751c9333476ecbfa6686ef409f596faf3 0.0s
=> => exporting config sha256:759d5c715f6e32bab56120f360ae3d85ed5d41b0ffb3fa0977 0.0s
=> => exporting attestation manifest sha256:e9ba78bfd6e2311625ce80f647d2828bd032 0.0s
=> => exporting manifest list sha256:540ec7659c6317cd09d7a835e351f605ce6d282f969 0.0s
=> => naming to docker.io/library/dashmarket-catalog:1.0          0.0s
=> => unpacking to docker.io/library/dashmarket-catalog:1.0       0.4s
PS C:\Users\User\Downloads\dashmarket-lab>
PS C:\Users\User\Downloads\dashmarket-lab> |
```



```

PS C:\Users\User\Downloads\dashmarket-lab> docker build -t dashmarket-order:1.0 ./order-
service
[+] Building 7.6s (10/10) FINISHED                                docker:desktop-linux
=> [internal] load build definition from Dockerfile                0.1s
=> => transferring dockerfile: 170B                                0.0s
=> [internal] load metadata for docker.io/library/node:lts-alpine 1.5s
=> [internal] load .dockerignore                                  0.0s
=> => transferring context: 2B                                       0.0s
=> [1/5] FROM docker.io/library/node:lts-alpine@sha256:d32cdf619f63fe0471182d089 0.0s
=> => resolve docker.io/library/node:lts-alpine@sha256:d32cdf619f63fe0471182d089 0.0s
=> [internal] load build context                                  1.7s
=> => transferring context: 2.30MB                                    1.7s
=> CACHED [2/5] WORKDIR /app                                       0.0s
=> [3/5] COPY package*.json ./                                    0.1s
=> [4/5] RUN npm install                                           3.0s
=> [5/5] COPY . .                                                  0.1s
=> exporting to image                                              0.9s
=> => exporting layers                                              0.3s
=> => exporting manifest sha256:9a625dd15d5994628c779f7052c4aa3c25376f0ef3dd3eb8 0.0s
=> => exporting config sha256:df7306ae0696f2158d0e9674c867321b7e28d317a510a9b895 0.0s
=> => exporting attestation manifest sha256:5abc1b9f40f8a07679f5c5ea188fccf3f4d4 0.0s
=> => exporting manifest list sha256:eed45c48d78a65ea7be03f6a60554cdaf9e40f48850 0.0s
=> => naming to docker.io/library/dashmarket-order:1.0            0.0s
=> => unpacking to docker.io/library/dashmarket-order:1.0         0.4s
PS C:\Users\User\Downloads\dashmarket-lab> |

```

```

PS C:\Users\User\Downloads\dashmarket-lab> docker build -t dashmarket-delivery:1.0 ./del
ivery-service
[+] Building 6.4s (10/10) FINISHED                                docker:desktop-linux
=> [internal] load build definition from Dockerfile                0.1s
=> => transferring dockerfile: 170B                                0.0s
=> [internal] load metadata for docker.io/library/node:lts-alpine 0.5s
=> [internal] load .dockerignore                                  0.1s
=> => transferring context: 2B                                       0.0s
=> [1/5] FROM docker.io/library/node:lts-alpine@sha256:d32cdf619f63fe0471182d089 0.1s
=> => resolve docker.io/library/node:lts-alpine@sha256:d32cdf619f63fe0471182d089 0.1s
=> [internal] load build context                                  1.4s
=> => transferring context: 2.30MB                                    1.4s
=> CACHED [2/5] WORKDIR /app                                       0.0s
=> [3/5] COPY package*.json ./                                    0.1s
=> [4/5] RUN npm install                                           3.2s
=> [5/5] COPY . .                                                  0.1s
=> exporting to image                                              0.8s
=> => exporting layers                                              0.3s
=> => exporting manifest sha256:1a706cc5c8ac917e351b3b382142c0a7a1049abd8b3fb41c 0.0s
=> => exporting config sha256:8c892b3b666fb633df471e9adca45ca2c4302e990722f02bda 0.0s
=> => exporting attestation manifest sha256:d1299de5434f112a8c47a3cc7318c3fe2845 0.0s
=> => exporting manifest list sha256:ea9bf533746a282cd5a75d96e10e2bc07509e968f0e 0.0s
=> => naming to docker.io/library/dashmarket-delivery:1.0        0.0s
=> => unpacking to docker.io/library/dashmarket-delivery:1.0     0.4s
PS C:\Users\User\Downloads\dashmarket-lab> |

```

```
PS C:\Users\User\Downloads\dashmarket-lab> docker build -t dashmarket-notify:1.0 ./notification-service
[+] Building 6.6s (10/10) FINISHED                                docker:desktop-linux
=> [internal] load build definition from Dockerfile                0.1s
=> => transferring dockerfile: 170B                               0.0s
=> [internal] load metadata for docker.io/library/node:lts-alpine 0.6s
=> [internal] load .dockerignore                                  0.0s
=> => transferring context: 2B                                     0.0s
=> [1/5] FROM docker.io/library/node:lts-alpine@sha256:d32cdf619f63fe0471182d089 0.0s
=> => resolve docker.io/library/node:lts-alpine@sha256:d32cdf619f63fe0471182d089 0.0s
=> [internal] load build context                                  1.3s
=> => transferring context: 2.30MB                                1.3s
=> CACHED [2/5] WORKDIR /app                                     0.0s
=> [3/5] COPY package*.json ./                                   0.1s
=> [4/5] RUN npm install                                         3.3s
=> [5/5] COPY . .                                               0.1s
=> exporting to image                                           0.9s
=> => exporting layers                                           0.3s
=> => exporting manifest sha256:a6add2b8f83ce2d4ede76f490c98b878bedefc094671b66b 0.0s
=> => exporting config sha256:0e1b010202797f2ecf532dbb3a15423e2b9ed0547c0d16ffd6 0.0s
=> => exporting attestation manifest sha256:4f1562cf1d980def4e1622cc210e58a55a10 0.0s
=> => exporting manifest list sha256:11daf73e3fe5f1d88e553808f878d67985ff35218c6 0.0s
=> => naming to docker.io/library/dashmarket-notify:1.0         0.0s
=> => unpacking to docker.io/library/dashmarket-notify:1.0      0.4s
PS C:\Users\User\Downloads\dashmarket-lab> |
```

```
PS C:\Users\User\Downloads\dashmarket-lab> docker images

```

IMAGE	ID	DISK USAGE	CONTENT SIZE	Info →	U	In Use
dashmarket-catalog:1.0	540ec7659c63	251MB	60.8MB			
dashmarket-delivery:1.0	ea9bf533746a	251MB	60.8MB			
dashmarket-notify:1.0	11daf73e3fe5	251MB	60.8MB			
dashmarket-order:1.0	eed45c48d78a	251MB	60.8MB			

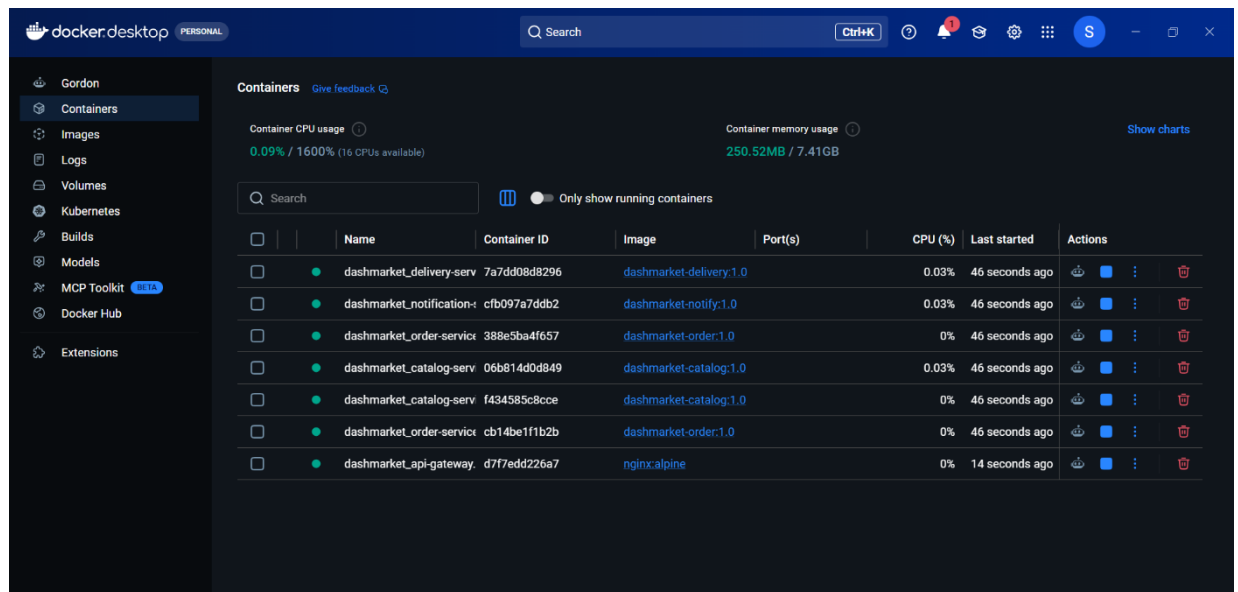
```
PS C:\Users\User\Downloads\dashmarket-lab> |
```

```
PS C:\Users\User\Downloads\dashmarket-lab> New-Item docker-compose.yml -ItemType File
```

Directory: C:\Users\User\Downloads\dashmarket-lab

Mode	LastWriteTime	Length	Name
-a----	8/23/2026 10:56 PM	0	docker-compose.yml

```
PS C:\Users\User\Downloads\dashmarket-lab> docker stack deploy --resolve-image=never -c
docker-compose.yml dashmarket
Since --detach=false was not specified, tasks will be created in the background.
In a future release, --detach=false will become the default.
Creating network dashmarket_dashmarket-net
Creating service dashmarket_api-gateway
Creating service dashmarket_catalog-service
Creating service dashmarket_order-service
Creating service dashmarket_delivery-service
Creating service dashmarket_notification-service
PS C:\Users\User\Downloads\dashmarket-lab> |
```



```
PS C:\Users\User\Downloads\dashmarket-lab> docker stack services dashmarket
ID                NAME                                MODE                REPLICAS    IMAGE
PORTS
uuict2fqtdcx     dashmarket_api-gateway            replicated          1/1         nginx:alpine
*:80->80/tcp
nkd29jdkqb6u     dashmarket_catalog-service        replicated          2/2         dashmarket-cata
log:1.0
ma9uu2nvwv7q     dashmarket_delivery-service        replicated          1/1         dashmarket-deli
very:1.0
cb614e2tp09v     dashmarket_notification-service    replicated          1/1         dashmarket-noti
fy:1.0
i20dx9cky08e     dashmarket_order-service           replicated          2/2         dashmarket-orde
r:1.0
PS C:\Users\User\Downloads\dashmarket-lab> |
```

Workspace

🔍

🔗

Invite

Upgrade

🔔

⚙️

🔄

—

□

✕

🔗 Overview

📁 My Collection

GET http://localhost/api/

+

⌵

No environment

⌵

☰

HTTP

http://localhost/api/catalog/items

📁 Save

⌵

Share

🔗

GET

⌵

http://localhost/api/catalog/items

G

⌵

☰ Docs

Params

Authorization

Headers 7

Body

Scripts

Tests

Settings

Cookies

Query Params

Key	Value	Description	Bulk Edit	...
Key	Value	Description		

Body

⌵

🔄

200 OK

• 38 ms

• 523 B

🌐

...

{ } JSON

⌵

▶ Preview

🖼️ Visualize

⌵

☰

☰

🔍

📄

🔗

1 {

2 "system": "DashMarket",

3 "service": "Catalog Service",

4 "data": [

5 {

6 "id": "101",

7 "name": "Organic Fresh Milk (1L)",

8 "category": "Dairy",

9 "price": 3.49

10 },

11 {

12 "id": "102",

13 "name": "Whole Grain Bread",

14 "category": "Bakery",

15 "price": 2.99

16 },

17 {

18 "id": "103",

19 "name": "Fresh Bananas (1kg)",

20 "category": "Produce",

21 "price": 1.89

22 }

23]

24 }

POST

http://localhost/api/notifications/notify



DocsParamsAuthorizationHeaders 8BodyScriptsTestsSettingsCookies

Query Params

Key	Value	Description	Bulk Edit	...
Key	Value	Description		

Body

200 OK • 19 ms • 282 B •  • ...

{ } JSON

PreviewVisualize

 |  |  |  | 

```
1 {
2   "message": "Notification received successfully"
3 }
```

Workspace

Invite

Upgrade

1

Overview

My Collection

POST http://localhost/api/

+

No environment

HTTP

http://localhost/api/orders/

Save

Share

POST

http://localhost/api/orders/

Send

Docs

Params

Authorization

Headers 8

Body

Scripts

Tests

Settings

Cookies

Query Params

Key	Value	Description	Bulk Edit	...
Key	Value	Description		

Body

201 Created

157 ms

327 B

...

{ } JSON

Preview

Visualize

1 {

2 "system": "DashMarket",

3 "service": "Order Service",

4 "message": "Order created successfully"

5 }

POST

http://localhost/api/notifications/notify

Send

Docs

Params

Authorization

Headers 9

Body

Scripts

Tests

Settings

Cookies

raw

JSON

Schema

Beautify

```
1  {
2    "event" : "Test_ALERT" ,
3    "message" : "Testing notification toute"
4  }
```

Body

200 OK • 16 ms • 282 B •

{ } JSON

Preview

Visualize

```
1  {
2    "message": "Notification received successfully"
3  }
```

POST ▼ Send ▼

Docs Params Authorization Headers 9 **Body** ● Scripts Tests Settings Cookies

raw ▼ JSON ▼ Schema Beautify

```

1  {
2    "itemId" : "101" ,
3    "message" : 1
4  }

```

Body ▼ 🔄 201 Created • 27 ms • 365 B • 🌐 | ⋮

{} JSON ▼ ▶ Preview 🖼️ Visualize ▼ 🔍 🔗 📄 🔗

```

1  {
2    "system": "DashMarket",
3    "service": "Order Service",
4    "message": "Order created successfully",
5    "order": {
6      "itemId": "101",
7      "message": 1
8    }
9  }

```

```

PS C:\Users\User\Downloads\dashmarket-lab> docker service logs dashmarket_notification-s
service
dashmarket_notification-service.1.wxn1gv5mua2e@docker-desktop |
dashmarket_notification-service.1.wxn1gv5mua2e@docker-desktop | > notification-servic
e@1.0.0 start
dashmarket_notification-service.1.wxn1gv5mua2e@docker-desktop | > node index.js
dashmarket_notification-service.1.wxn1gv5mua2e@docker-desktop |
dashmarket_notification-service.1.wxn1gv5mua2e@docker-desktop | DashMarket Notificati
on Service running on port 5004
dashmarket_notification-service.1.wxn1gv5mua2e@docker-desktop | [2026-08-23T17:40:09.
482Z] Notification received: undefined
dashmarket_notification-service.1.wxn1gv5mua2e@docker-desktop | [2026-08-23T17:41:08.
561Z] Notification received: { message: 'New order received' }
dashmarket_notification-service.1.wxn1gv5mua2e@docker-desktop | [2026-08-23T17:47:14.
913Z] Notification received: { event: 'Test_ALERT', message: 'Testing notification toute
' }
dashmarket_notification-service.1.wxn1gv5mua2e@docker-desktop | [2026-08-23T17:48:13.
165Z] Notification received: {
dashmarket_notification-service.1.wxn1gv5mua2e@docker-desktop |   message: 'New order
received',
dashmarket_notification-service.1.wxn1gv5mua2e@docker-desktop |   order: { event: 'Te
st_ALERT', message: 'Testing notification toute' }
dashmarket_notification-service.1.wxn1gv5mua2e@docker-desktop | }
dashmarket_notification-service.1.wxn1gv5mua2e@docker-desktop | [2026-08-23T17:49:19.
532Z] Notification received: { message: 'New order received', order: { itemId: '101', me
ssage: 1 } }

```